

DP014
Physics 1
Semester I
Session 2024/2025
2 hours

DP014
Fizik 1
Semester I
Sesi 2024/2025
2 jam



KEMENTERIAN PENDIDIKAN

BAHAGIAN MATRIKULASI
MATRICULATION DIVISION

PRA PEPERIKSAAN SEMESTER PROGRAM MATRIKULASI
MATRICULATION PROGRAMME EXAMINATION

JANGAN BUKA KERTAS SOALAN INI SEHINGGA DIBERITAHU.
DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO.

Kertas soalan ini mengandungi **11** halaman bercetak.

*This question paper consists of **11** printed pages.*

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SULIT

SULIT

INSTRUCTIONS TO CANDIDATE:

This question paper consists of **7** questions.

Answer **all** questions.

All answers must be written in the answer booklet provided. Use a new page for each question.

The use of non-programmable scientific calculator is permitted.

ARAHAN KEPADA CALON:

Kertas soalan ini mengandungi 7 soalan.

*Jawab **semua** soalan.*

Semua jawapan hendaklah ditulis pada buku jawapan yang disediakan. Gunakan muka surat baharu bagi nombor soalan yang berbeza.

Penggunaan kalkulator saintifik yang tidak boleh diprogramkan dibenarkan.

SULIT

LIST OF SELECTED CONSTANT VALUES
SENARAI NILAI PEMALAR TERPILIH

Speed of light in a vacuum <i>Laju cahaya dalam vakum</i>	c	$= 3.00 \times 10^8 \text{ m s}^{-1}$
Permeability of free space <i>Ketelapan ruang bebas</i>	μ_0	$= 4\pi \times 10^{-7} \text{ H m}^{-1}$
Permittivity of free space <i>Ketelusan ruang bebas</i>	ϵ_0	$= 8.85 \times 10^{-12} \text{ F m}^{-1}$
Electron charge magnitude <i>Magnitud cas elektron</i>	e	$= 1.60 \times 10^{-19} \text{ C}$
Planck constant <i>Pemalar Planck</i>	h	$= 6.63 \times 10^{-34} \text{ J s}$
Electron mass <i>Jisim elektron</i>	m_e	$= 9.11 \times 10^{-31} \text{ kg}$ $= 5.49 \times 10^{-4} \text{ u}$
Neutron mass <i>Jisim neutron</i>	m_n	$= 1.674 \times 10^{-27} \text{ kg}$ $= 1.008665 \text{ u}$
Proton mass <i>Jisim proton</i>	m_p	$= 1.672 \times 10^{-27} \text{ kg}$ $= 1.007277 \text{ u}$
Deuteron mass <i>Jisim deuteron</i>	m_d	$= 3.34 \times 10^{-27} \text{ kg}$ $= 2.014102 \text{ u}$
Molar gas constant <i>Pemalar gas molar</i>	R	$= 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Avogadro constant <i>Pemalar Avogadro</i>	N_A	$= 6.02 \times 10^{23} \text{ mol}^{-1}$
Boltzmann constant <i>Pemalar Boltzmann</i>	k	$= 1.38 \times 10^{-23} \text{ J K}^{-1}$
Free-fall acceleration <i>Pecutan jatuh bebas</i>	g	$= 9.81 \text{ m s}^{-2}$
Atomic mass unit	1 u	$= 1.66 \times 10^{-27} \text{ kg}$
<i>Unit jisim atom</i>		$= 931.5 \frac{\text{MeV}}{c^2}$

SULIT

Electron volt <i>Elektron volt</i>	1 eV	$= 1.6 \times 10^{-19} \text{ J}$
Constant of proportionality for Coulomb's law <i>Pemalar hukum Coulomb</i>	$k = \frac{1}{4\pi\epsilon_0}$	$= 9.0 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$
Atmospheric pressure <i>Tekanan atmosfera</i>	1 atm	$= 1.013 \times 10^5 \text{ Pa}$
Density of water <i>Ketumpatan air</i>	ρ_w	$= 1000 \text{ kg m}^{-3}$

LIST OF SELECTED FORMULAE
SENARAI RUMUS TERPILIH

1. $v = u + at$	19. $s = r\theta$
2. $s = ut + \frac{1}{2}at^2$	20. $v = r\omega$
3. $v^2 = u^2 + 2as$	21. $a_t = r\alpha$
4. $s = \frac{1}{2}(u + v)t$	22. $\omega = \omega_o + \alpha t$
5. $v = u - gt$	23. $\theta = \omega_o t + \frac{1}{2}\alpha t^2$
6. $v^2 = u^2 - 2gs$	24. $\omega^2 = \omega_o^2 + 2\alpha\theta$
7. $s = ut - \frac{1}{2}gt^2$	25. $\theta = \frac{1}{2}(\omega_o + \omega)t$
8. $p = mv$	26. $\tau = rF\sin\theta$
9. $J = F\Delta t$	27. $\frac{dQ}{dt} = -kA\left(\frac{\Delta T}{L}\right)$
10. $J = \Delta p = mv - mu$	28. $v_{rms} = \sqrt{\langle v^2 \rangle}$
11. $f_s \leq \mu_s N$	29. $v_{rms} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3RT}{M}}$
12. $f_k = \mu_k N$	30. $\Delta U = Q - W$
13. $W = \vec{F} \cdot \vec{s} = F\cos\theta$	
14. $K = \frac{1}{2}mv^2$	
15. $U = mgh$	
16. $U_s = \frac{1}{2}kx^2 = \frac{1}{2}Fx$	
17. $a_c = \frac{v^2}{r} = r\omega^2 = v\omega$	
18. $F_c = \frac{mv^2}{r} = mr\omega^2 = mv\omega$	

SULIT

- 1 (a) Calculate the density of a solid cube that measures 5.0 cm on each side and has a mass of 350 g in SI unit.

[3 marks]

- (b) A 3.0 N force acts on an object in the positive x -direction, and a 5.0 N force acts on the same object at an angle of 60° above the positive x -axis. Determine the magnitude and direction of the resultant force.

[5 marks]

- 2 (a) A car starts from rest at a point **A** on a straight racetrack. The car moves with constant acceleration for 20 s, reaching a speed of 25 m s^{-1} . The car then travels at a constant speed of 25 m s^{-1} for 120 s. Finally, it moves with constant deceleration, coming to rest at a point **B**. Total distance between **A** and **B** is $4.0 \times 10^3 \text{ m}$.

(i) Sketch a speed – time graph for the motion of the car.

(ii) Calculate the total time the car takes to travel from **A** to **B**.

[5 marks]

- (b) A racing car moves with constant acceleration along a straight horizontal road. It passes the point **O** with speed 12 m s^{-1} . It passes the point **A**, 4 seconds later with speed 60 m s^{-1} .

(i) Calculate the acceleration of the car.

(ii) Determine the distance from point **O** to **A**.

[4 marks]

- (c) A construction worker accidentally drops a brick from a high scaffold. Calculate the velocity of the brick after 4.0 s. Explain your answer.

[3 mark]

SULIT

- 3 (a) A force with magnitude of 540 N is used to stop an object with a mass of 65 kg moving at a velocity of 175 m s^{-1} right. How long will it take to bring the object to a full stop?

[2 marks]

- (b) A 7.0 kg ball moving at 0.60 m s^{-1} collides with a 4.5 kg ball at rest. If the heavier ball moves at 0.30 m s^{-1} after the collision, how fast does the lighter ball move?

[2 marks]

- (c) A 55 kg ball is moving at 65 m s^{-1} to the right when it collides with a stationary 24 kg ball. After the collision, the 24 kg ball is moving at 85 m s^{-1} , 20° above the right. What is the final velocity of the 55 kg ball?

[5 marks]

4 (a)

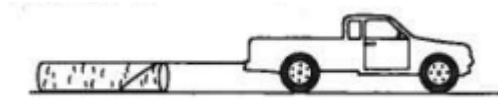
**FIGURE 4**

FIGURE 4 shows a 1150 kg car pulls a 450 kg wood using a rope. The car exerts a horizontal force of 3.8×10^3 N against the ground in order to accelerate. The coefficient friction between the wood to the ground is 0.15.

(i) Calculate the acceleration of the system.

(ii) Calculate tension in the rope.

[5 marks]

(b) A 50 kg box is dragged on a horizontal floor through a distance of 1.5 m by a 300 N force at 30° above the horizontal. The coefficient of friction of the floor is 0.2.

(i) Sketch a labelled diagram showing **all** the forces on the box.

(ii) Calculate the normal force.

[7 marks]

SULIT

- 5 (a) A rubber ball moves in a circular path of radius 100 cm. It makes two revolutions per second. Calculate the

(i) period of revolution.

(ii) speed of the ball.

[4 marks]

- (b) A 900 kg race car makes one lap around a track of radius 50.0 m in 12.0 s.

(i) Calculate the car's angular speed.

(ii) Calculate the centripetal force acting on the car.

(iii) State the force that acts as a centripetal force in this system.

[5 marks]

- 6 (a) As you drive down the road at 17 m s^{-1} , you press on the gas pedal and speed up with a uniform acceleration of 1.12 m s^{-2} for 0.65 s . If the tires on your car have a radius of 33 cm , calculate the
- (i) angular velocity.
 - (ii) angular acceleration.
 - (iii) angular displacement.
 - (iv) number of revolutions during the 0.65 s interval.

[8 marks]

(b)

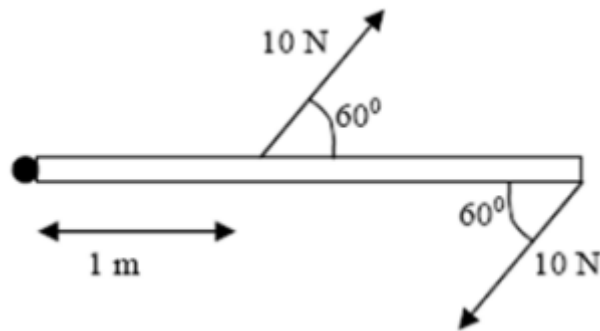


FIGURE 6

FIGURE 6 shows the top view of a door that is 2 m wide. Two forces are applied to the door as indicated in the diagram. Calculate the net torque on the door with respect to the hinge. Explain your answer.

[4 marks]

- 7 (a) A perfectly insulated aluminium rod has length 50 cm and cross – sectional area 3.0 cm^2 . At the steady state at both ends of the rod are 150°C and 50°C respectively.

[Thermal conductivity of aluminium is $210 \text{ W m}^{-1} \text{ K}^{-1}$]

- (i) Sketch a labelled graph of temperature against distance.
- (ii) Calculate the temperature gradient along the rod.
- (iii) Calculate the rate of heat flow in the rod.

[7 marks]

- (b) A gas tank containing nitrogen gas at a temperature of 27°C is heated to 352°C . Given that the molar mass of nitrogen is 28 g mol^{-1} , calculate the change in root mean square (v_{rms}) of the nitrogen gas molecules.

[5 marks]

- (c) An ideal gas is compressed isothermally from $4V$ to V . The work done on the gas is $4.5 \times 10^3 \text{ J}$.

- (i) Sketch a P - V graph for this process.
- (ii) Calculate the heat transferred during the compression. Is the heat absorbed or released by the system?

[6 marks]

END OF QUESTION PAPER